

COMSATS UNIVERSITY ISLAMABAD

ATTOCK CAMPUS



Project Proposal

for

Smart Travel Planner

by

Malaika Wasiq

Amima Masood

CIIT/SP23-BSE-041/ATK

CIIT/SP23-BSE-038/ATK

Supervisor

Mr. Muhammad Waseem Khan

Bachelor of Science in Software Engineering (2023–2026)

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Amima Masood

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As the supervisor, I have reviewed the proposed project and found it to be feasible, relevant, and aligned with the academic and industry standards. I recommend this project for evaluation by the committee to assess its viability and approval for further development.

Supervisor Signature:_____

Mr. Muhammad Waseem Khan

Department of Computer Science, (CUI) Attock Campus

Project Summary

Travel planning is often seen as a difficult task because different websites and apps must be visited to check weather, hotels, tourist places, maps, and budgets. For many travelers, especially students and budget-conscious people, this process feels complicated and time-consuming. This study has been proposed to make trip planning easier and more enjoyable with the help of artificial intelligence.

The main challenge in this area is that most travel apps provide general recommendations. These apps do not adjust plans according to personal interests, budgets, or sudden changes such as weather. Some applications focus only on hotel bookings or basic travel guides, but very few offer an adaptive and personalized experience.

This gap highlights the need for a system that can think beyond static plans and create flexible itineraries. In this project, the problem is defined as the absence of a free and intelligent solution that can generate personalized travel plans. The objective is to design a mobile application that uses AI to build day-wise itineraries based on the user's destination, budget, days, and interests. The methodology includes taking user inputs, connecting with external APIs for weather, maps, and tourist information, and then generating dynamic plans stored in a database for future use.

The proposed work is expected to save time, reduce stress, and provide a smarter travel experience. By giving importance to real-time updates and personal preferences, this system can bring a more human touch to travel planning. It will allow people to focus on enjoying their journey instead of worrying about arrangements.

1 Introduction

Artificial intelligence (AI) has recently changed the way that digital applications support individuals in their daily lives. One area that is expanding is travel planning, which can be overwhelming for many individuals. There are many aspects of travel planning, including people needing to go to different platforms to check things out; for example, checking the weather, finding tourist attractions, finding a hotel, and tracking budgets. Multiple mobile applications exist to help, but they typically return basic results and do not meet the user's individual need or preferences. Therefore, the proposed project is to design and develop an Mobile application that will produce intelligent, day-wise travel itineraries. The application will consist of visual and actual user input data, such as destination, budget, and interests; and external source data, such as weather data, maps, and tourist information. The goal of the app is to create user traveler plans quickly while eliminating manual searches on outside platforms. The system will utilize AI models for generating recommendations, external data sources (e.g., Expected Weather and Location) to provide real-time weather and location data, and a solo interactive map for guidance and road management. A database will store trip details and monitor user preferences and budgets so that they can be re-used or updated if the user desires. Instead of providing a static travel guide, & the focus on this project is based on adaptability and personalization for the user experience. The travel itinerary will adjust on its own if it rains or the user's budget level is exceeded; both of which impacted the user's experience. In addition, features built into the app include easy budget tracking, boosting hotel management, notifications once the budget is reached or exceeded, and an option to export itineraries.

2 Literature Review

Several travel planning applications are already available in the market, and they provide different features to assist travelers. However, most of these systems have noticeable limitations in terms of personalization, adaptability, and budget sensitivity. A detailed review of three popular applications, namely TripIt, Roadtrippers, and Flaia (AI), is presented below.

TripIt [1] is a very popular travel planning app. It automatically collects bookings like flights, hotels, and car rentals, and puts them together in one plan. This is very useful for travelers who want all their travel details in one place. However, TripIt mainly works as an organizer, not a planner. It cannot create daily travel schedules or adjust to sudden changes like bad weather, traffic, or budget limits. So, while it is good for managing reservations, it does not provide flexible or personalized planning.

Roadtrippers [2] is another well-known app, mostly for road trips. It lets users pick a route and then suggests attractions, restaurants, and other spots along the way. This is helpful for long drives and cross-country trips. But its use is limited since it only focuses on road travel. It also does not handle flights, hotel bookings, or real-time updates. In addition, it does not include budget tracking, which makes it less useful for travelers who need to control their expenses.

Flaia (AI) [3] is a more modern app that uses artificial intelligence to give personalized suggestions. It recommends attractions, builds itineraries, and tries to match them with the user’s interests. Even though Flaia is more advanced than traditional systems, it still has some big gaps. It has very little support for budget planning, and it does not fully connect with live data like weather or hotel availability. Also, it cannot update plans automatically when travel conditions suddenly change.

From this review, it is clear that existing apps have useful features but none of them provide a complete and flexible solution. The proposed project aims to fill this gap by creating an AI-powered mobile app that makes personalized, day-wise travel plans. Unlike current systems, it will combine live weather updates, maps, tourist information, hotel suggestions, and budget tracking into one platform. The system will also update plans in real time, making it more reliable and user-friendly. A comparison of these apps with the proposed system is shown in Table 1.

Table 1: Related System Analysis with Proposed Solution

Application Name	Weakness	Proposed Project Solution
TripIt [1]	Organizes bookings but does not create day-wise itineraries, lacks real-time adaptability and budget consideration.	Generates personalized, day-wise itineraries that adapt to real-time weather, user interests, and budget.
Roadtrippers [2]	Limited to road trips; lacks integration for flights, hotels, and budget optimization.	Provides a complete travel planning solution covering attractions, hotels, maps, routes, and budget tracking.
Flaia (AI) [3]	Uses AI but has weak budget control, limited integration with live data, and lacks dynamic adaptability.	Employs AI with real-time weather, maps, tourist spot, and hotel data to generate adaptive and optimized travel plans.

3 Problem Statement

Today, there is so much online information about travel destinations, hotels, routes, and tourist spots that planning a trip has become more difficult. Current apps like Roadtrippers, TripIt, and Flaia (AI) only give partial solutions and do not fully meet user needs. Roadtrippers is limited to road trips, TripIt mainly focuses on reservations, and Flaia has weak budget management and little real-time flexibility. Because of this, travelers often face problems in making effective and customized plans that balance cost, convenience, and personal preferences.

Most existing apps also do not provide complete real-time updates such as traffic conditions, hotel availability, or weather changes. This makes their plans less reliable, especially when unexpected situations occur during travel. Another problem is that many of these apps lack proper cost-control features, so travelers with limited budgets struggle to manage expenses.

Therefore, there is a clear need for a smart and flexible system that can create full day-by-day itineraries while considering user preferences, financial limits, and real-time conditions. This project aims to fill this gap by developing a mobile app powered by AI that includes live weather, maps, hotels, tourist attractions, and budget tracking. Compared to current tools, the proposed solution will give travelers a more personalized, reliable, and affordable planning experience.

4 Proposed Solution

The proposed solution is an AI-powered, user-friendly, and flexible travel planning system that addresses the weaknesses of current apps. It can be explained in the following points:

- An intelligent itinerary generator will create personalized day-by-day travel plans based on user details such as destination, time, budget, and interests.
- Real-time data will be used to keep plans updated with live information like weather, traffic, and hotel availability.
- A budget management system will help reduce costs by suggesting affordable routes, hotels, and attractions while staying within the user's budget.
- A recommendation system will suggest popular tourist spots, restaurants, and activities using AI-based decision-making.

- The app will have an easy-to-use interface where travelers can modify plans, add or remove destinations, and customize schedules.
- Cloud storage will be used so that itineraries can be accessed on multiple devices, making it more convenient.
- Security and privacy measures will be added to protect user data such as travel history, budget details, and personal information.

5 Vision Statement

This project aims to change the way people plan and enjoy their trips by offering a smart, flexible, and easy-to-use travel companion. The app will provide personalized itineraries, real-time updates, and cost-saving suggestions to make travel planning easier with the help of AI. The goal is to build a platform that not only increases convenience but also helps travelers have better and smarter experiences by fixing the problems found in current travel apps. In the end, the objective is to give travelers a reliable personal assistant that ensures stress-free, enjoyable, and well-organized trips tailored to their own preferences.

6 Scope

The scope of this project covers the design and development of a mobile-based Smart Travel Planner application that provides personalized and adaptive travel itineraries. The application will operate entirely on free-tier services and openly available datasets to ensure accessibility without additional costs. Users will be able to input destination, budget, travel days, and personal interests, and the system will generate a day-wise itinerary accordingly.

The scope includes the integration of:

- **OpenWeather API [4]** for live weather updates and forecasts.
- **Wikipedia API [5]** for descriptions and images of tourist attractions.
- **OpenStreetMap and OpenRouteService APIs [6]** for maps, routes, and navigation.
- **HuggingFace Inference API [7]** for AI-driven recommendations and itinerary generation.

- **MongoDB Atlas [8]** for secure cloud-based storage of user data and trip history.
- **React Native (Frontend)** and **Node.js + Express (Backend)** as core development frameworks [9].

The application will provide modules for hotel management (through static dataset or free APIs), budget tracking, weather alerts, notification services, and trip storage.

The scope is limited to free services and free-tier APIs only, no premium booking APIs, flight reservations, or paid third-party integrations will be implemented in this phase. Future enhancements may extend toward real-time bookings, multilingual support, chatbot-based assistance, and AR-based travel guidance.

7 Methodology

The methodology of the proposed system is based on a systematic and modular approach to ensure efficiency, adaptability, and reliability of the Smart Travel Planner application. The development follows step-by-step procedures combined with free-tier APIs and AI models for generating personalized travel itineraries. The methodology is described as follows:

1. **Requirement Gathering:** User requirements such as destination, travel dates, budget, and interests are collected through the mobile application interface.
2. **Pre-processing:** Input is validated, cleaned, and structured to make it compatible with the AI-based itinerary generator.
3. **AI Itinerary Generation:** The HuggingFace inference model processes user inputs to generate a personalized, day-wise travel plan.
4. **Integration with APIs:**
 - OpenWeather API provides real-time weather data to adjust plans dynamically.
 - Wikipedia API supplies descriptions and images of tourist attractions.
 - OpenStreetMap and OpenRouteService APIs provide maps, routes, and navigation assistance.
5. **Budget Tracking:** A cost-estimation module calculates expenses for hotels, transport, food, and activities, ensuring the trip remains within the budget.

6. **Hotel & Storage Modules:** Hotels are suggested using datasets/free APIs, and all travel data is stored in MongoDB Atlas (cloud).
7. **Notifications:** Alerts are triggered for weather changes, upcoming trips, and reminders using push notification services.
8. **Output:** A complete, day-wise personalized itinerary is presented with maps, cost breakdown, and options to save, export, or share.

The system development will follow an iterative and modular design, ensuring flexibility for enhancements such as chatbot-based guidance, AR navigation, and live booking in the future.

The step-by-step flow of the proposed system is shown in Figure 1.

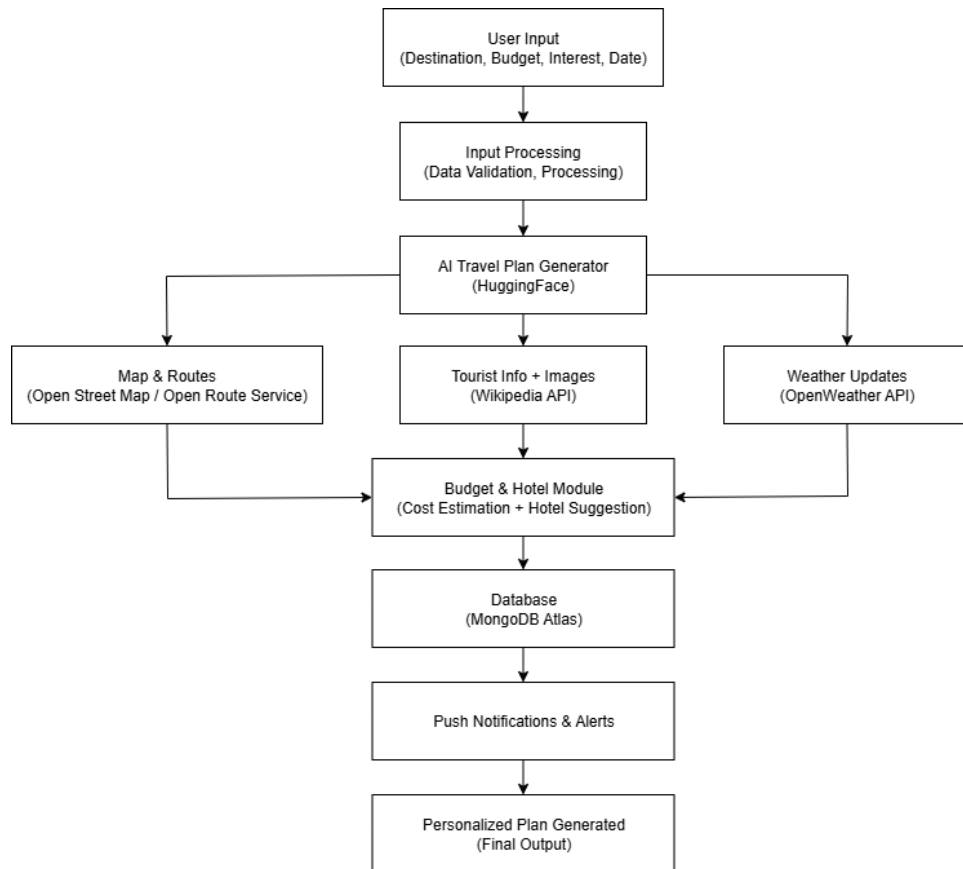


Figure 1: Block Diagram of Proposed Solution

8 Tools and Technologies

The development of the proposed solution is based on modern tools and technologies that are freely available or supported through free-tier services. The tools and technologies used in proposed solution are listed as follows:

- **Frontend:** React Native framework is employed for cross-platform mobile application development, ensuring compatibility with both Android and iOS devices.
- **Backend:** Node.js with Express framework is utilized to build the server-side application, providing APIs for communication between frontend and backend modules.
- **Database:** MongoDB Atlas (Free Tier) is used for cloud storage of user profiles, trips, and history.
- **Artificial Intelligence:** HuggingFace Inference API (Free Tier) is integrated to generate personalized itineraries.
- **Weather API:** OpenWeather API (Free Tier) is employed to fetch live weather conditions and forecasts.
- **Tourist Information API:** Wikipedia API is used for retrieving descriptions, images, and information about tourist attractions.
- **Maps and Navigation:** OpenStreetMap and OpenRouteService APIs are applied to display maps, routes, and calculate travel times.
- **Notifications:** Free notification services and APIs are integrated to provide reminders, weather alerts, and upcoming trip notifications.
- **Development Environment:** Visual Studio Code is used as the primary integrated development environment (IDE), while GitHub is used for version control and collaborative development.

9 Applications of the Proposed Solution

This proposed solution will have a wide range of applications, including:

- **Personal Travel Assistance:** The application will be used by individuals to generate personalized travel itineraries that will match their budget, time, and interests.
- **Student and Budget Travelers:** It will assist students and budget-conscious travelers by suggesting cost-effective plans and accommodations through free-tier data sources.
- **Tourism Promotion:** Local tourism authorities will benefit from this system by promoting tourist destinations with AI-generated itineraries.
- **Travel Agencies:** Travel companies will be able to adopt the system as a digital tool to provide smart itineraries and enhance customer engagement without manual effort.
- **Educational Purpose:** The project will serve as a learning model for students and researchers working in the field of artificial intelligence, mobile development, and API integration.
- **Emergency and Adaptive Travel Planning:** In case of sudden weather changes or time constraints, the system will quickly update travel plans to ensure safety and efficiency.

10 WBS and Gantt Chart

The Work Breakdown Structure (WBS) is defined as a hierarchical decomposition of the total work into smaller, more manageable sections. It will help in dividing the project into phases, tasks, and deliverables that will be completed step by step [10]. On the other hand, the Gantt Chart is defined as a visual timeline that will show the planned schedule of tasks and their dependencies throughout the development of the project. Both tools will ensure that the project will be managed effectively and progress will be tracked in an organized way [11].

10.1 Work Breakdown Structure (WBS)

The following hierarchical structure represents the planned activities for the proposed solution:

1. Project Initiation

- Requirement gathering
- Analysis

2. System Design

- UI Design
- Frontend Development

3. Implementation

- Backend development
- Database Development

4. Testing

- Unit testing
- Integration testing
- User acceptance testing

5. Maintenance and Future Enhancements

- Bug fixing
- Adding premium APIs
- Improving AI recommendations

10.2 Gantt Chart

The Gantt Chart will represent the schedule of project tasks along a timeline [11]. Each major activity will be assigned specific weeks to ensure that the project will be completed in time. The planned schedule is given below:

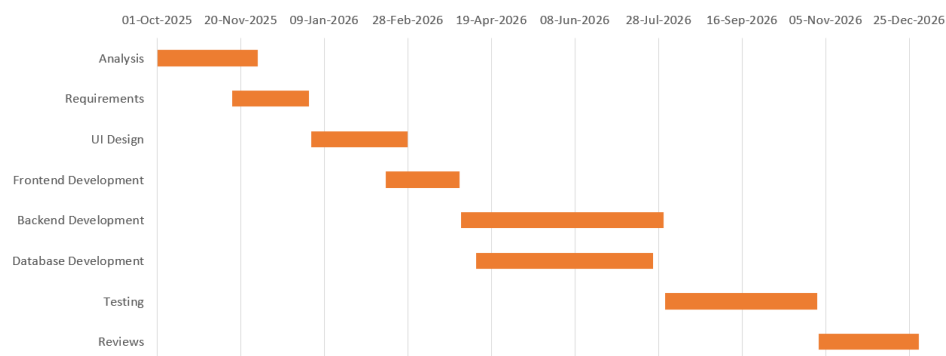


Figure 2: Gantt Chart

11 Mockups

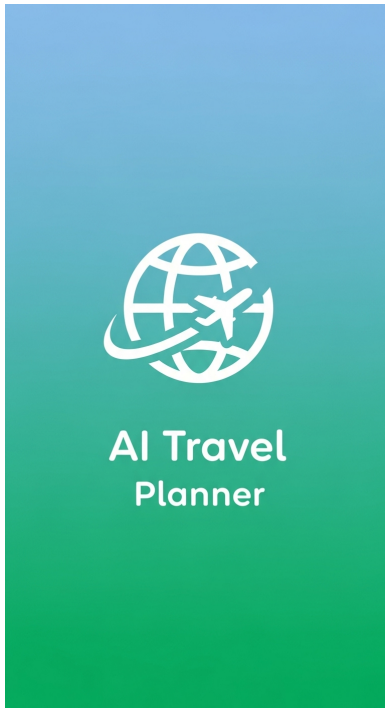


Figure 3: Splash Screen

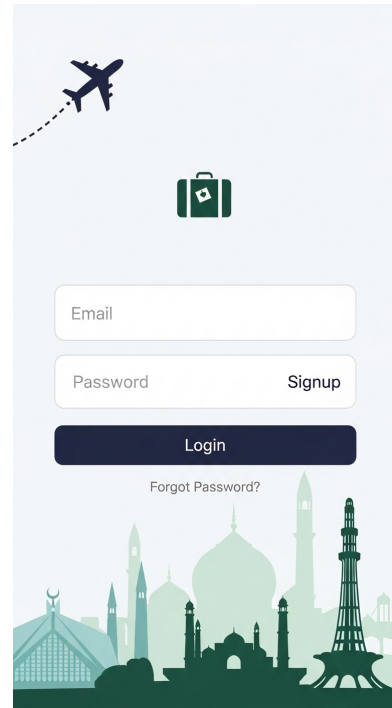


Figure 4: Login Screen

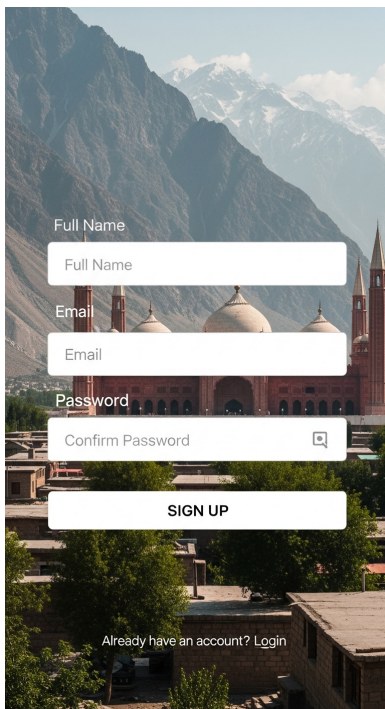


Figure 5: Signup Screen

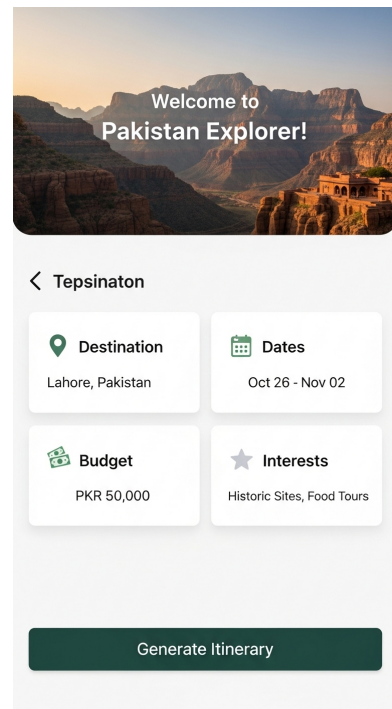


Figure 6: Home Screen

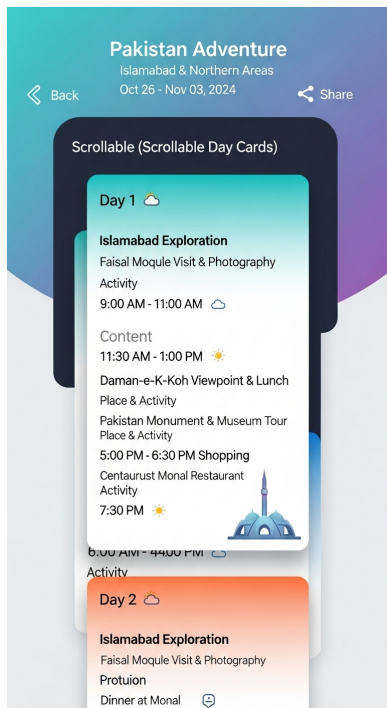


Figure 7: Itinerary Screen

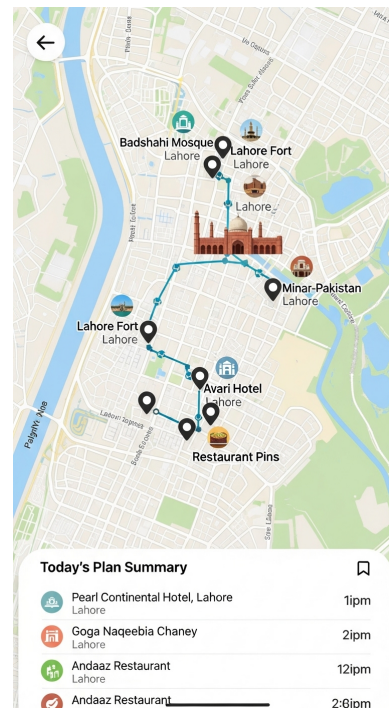


Figure 8: Map View Screen

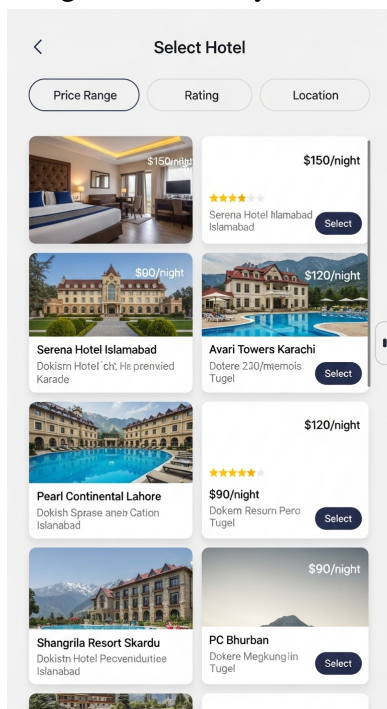


Figure 9: Hotel Selection Screen

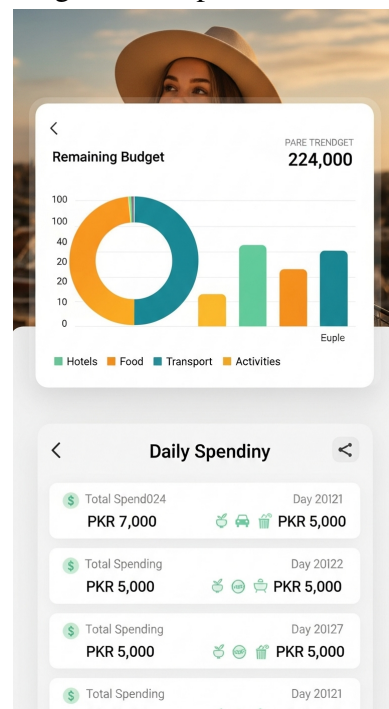


Figure 10: Budget Tracker Screen

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